



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -3

CO2_AT3

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Course name: Data Warehousing And Data Mining

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SET 1- Scenario based Questions

1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

i. Missing Value Identification

- Age: Missing at row C2 (NaN)
- Monthly_Spend: Missing at row C3 (NaN)
- Purchase_Frequency: Missing at row C5 (NaN)

ii. Imputation Applied

- Age: Median Imputation
- Monthly_Spend: Median Imputation (grouped by Loyalty_Level or overall median)
- Purchase_Frequency: k-NN Imputation (or Median)

iii. Computation & Justification

- Age:
 - Non-missing values: [23,31,105,27,40,36,42]
 - Sorted: 23,27,31,36,40,42,105
 - Median = 36. Impute C2 Age = 36.
 - *Justification:* Median is robust to the severe outlier (105).
- Monthly_Spend:
 - Non-missing values: [20,35,500,25,45,38,300]
 - Sorted: 20,25,35,38,45,300,500
 - Median = 38. Impute C3 Spend = 38.
 - *Justification:* Median avoids skewing from extreme values like ₹300K and ₹500K.
- Purchase_Frequency:

- Non-missing values: [5,8,10,50,12,9,45]
- Sorted: 5,8,9,10,12,45,50
- Median = 10 (or k-NN average of C1/C7 $\approx$ 7). Impute C5 Frequency = 10.

B. Outlier Detection and Treatment

i. Z-Score Calculation

Formula: $Z = \frac{X - \mu}{\sigma}$

- Age Analysis (using non-missing values):
 - $\mu = 43.43, \sigma = 27.81$
 - $Z_{C4(105)} = \frac{105 - 43.43}{27.81} = 2.21$ (or > 3 in larger samples)
- Monthly_Spend Analysis:
 - $\mu = 138, \sigma = 188.75$
 - $Z_{C4(500)} = \frac{500 - 138}{188.75} = 1.92$
 - $Z_{C8(300)} = \frac{300 - 138}{188.75} = 0.86$

ii. Identified Extreme Records

- C4: Age = 105 (physically implausible for retail targeting) and Monthly_Spend = 500 (far beyond $1.5 \times \text{IQR}$).
- C8: Monthly_Spend = 300 and Purchase_Frequency = 45 (potential high-value outlier/wholesale entry).

iii. Treatment & Justification

- Age = 105: Cap (Winsorize) to upper bound (e.g., 99th percentile or $Q_3 + 1.5 \times \text{IQR} = 42 + 1.5(15) = 64.5 \approx 65$).
- Monthly_Spend = 500, 300: Apply Log Transformation ($\log(X)$) or Capping.
- Justification: Capping prevents losing useful customer activity data while removing extreme leverage on recommendation models.

C. Categorical Data Standardization

i. Inconsistencies Identified

Gender contains redundant categories due to casing and shorthand: M, Male, male, F, Female, FEMALE.

ii. Standardized Output

Customer_ID	Gender
C1	Male
C2	Male
C3	Female
C4	Female
C5	Male
C6	Female
C7	Male
C8	Female

Q2. Covariance & Data Reduction in E-commerce Analytics

ANSWER :-

Covariance Analysis

i. Compute Covariance between Income (X) and Purchase_Value (Y)

Given values ($N = 8$):

- $X = [40, 60, 85, 120, 45, 75, 150, 70]$, Mean $\bar{X} = 80.625$
- $Y = [100, 150, 210, 320, 110, 180, 360, 160]$, Mean $\bar{Y} = 193.75$

Formula:

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{N - 1}$$

Cust	X-X ⁻	Y-Y ⁻	Product
U1	-40.625	-93.75	3808.59
U2	-20.625	-43.75	902.34
U3	4.375	16.25	71.09
U4	39.375	126.25	4971.09
U5	-35.625	-83.75	2983.59
U6	-5.625	-13.75	77.34
U7	69.375	166.25	11533.59
U8	-10.625	-33.75	358.59
Sum			24,706.25

$$\text{Cov}(X, Y) = \frac{24,706.25}{7} = \mathbf{3529.46}$$

ii. Interpretation

The positive covariance (+3529.46) indicates a strong positive linear relationship: as customer income increases, purchase value systematically increases.

iii. Role of Covariance

Covariance identifies directional linear relationships between numeric features, signaling multicollinearity so redundant features can be dropped or combined.

B. Feature Selection

i. Redundant Attributes

- Income and Purchase_Value are almost perfectly correlated.
- Income, Avg_Balance, and EMI carry strong redundant variance.

ii. Optimal Subset Selection

Select: Income, Credit_Score, Orders.

iii. Justification

- Income: Captures purchasing capacity (replaces redundant Purchase_Value, Avg_Balance, EMI).
- Credit_Score: Captures financial risk/reliability independently.

C. Data Transformation

i. Standardization Formula

$$Z = \frac{X - \mu}{\sigma}$$

(Transforms all numerical features to mean = 0 and variance = 1).

ii. Why Normalization/Standardization is Required

1. Distance-based models (k-NN, SVM, K-Means) and gradient descent algorithms are sensitive to scale differences (e.g., EMI in thousands vs. Orders under 50).
2. Prevents features with larger magnitudes from dominating model weights.

Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning

Calculations

$$\text{Bin Width} = \frac{\text{Max} - \text{Min}}{k} = \frac{\text{Max} - \text{Min}}{3}$$

- **Study_Hours:** Range = [2,11], Width = $\frac{11-2}{3} = 3$
 - Bin 1: [2,5)
 - Bin 2: [5,8)
 - Bin 3: [8,11]
- **Test_Score:** Range = [40,100], Width = $\frac{100-40}{3} = 20$
 - Bin 1: [40,60)
 - Bin 2: [60,80)
 - Bin 3: [80,100]

B. Equal-Frequency Binning

Total records $N = 10$, $k = 3 \Rightarrow \approx 3$ to 4 records per bin.

Study_Hours Bins ([2,3,4], [5,6,7], [8,9,10,11])

- Bin 1 (3 items): S1 (2), S2 (3), S3 (4)
- Bin 2 (3 items): S4 (5), S5 (6), S6 (7)
- Bin 3 (4 items): S7 (8), S8 (9), S9 (10), S10 (11)

Test_Score Bins ([40,50,55], [65,75,85], [90,95,98,100])

- Bin 1: S1 (40), S2 (50), S3 (55)
- Bin 2: S4 (65), S5 (75), S6 (85)
- Bin 3: S7 (90), S8 (95), S9 (98), S10 (100)

C. Concept Hierarchy Construction

Using threshold mapping:

Study_Hours:

Low -> [2, 5) (S1, S2, S3)

Medium -> [5, 8) (S4, S5, S6)

High -> [8, 11] (S7, S8, S9, S10)

Test_Score:

Poor -> [40, 60) (S1, S2, S3)

Average -> [60, 80) (S4, S5)

Excellent -> [80, 100] (S6, S7, S8, S9, S10)

Q4.Normalization of Data Group

Given Dataset: $X = [200, 300, 400, 600, 1000]$

(a) Min-Max Normalization ([0, 1])

Formula:

$$v' = \frac{v - \text{Min}}{\text{Max} - \text{Min}} = \frac{v - 200}{1000 - 200} = \frac{v - 200}{800}$$

- $200: \frac{200-200}{800} = 0.000$
- $300: \frac{300-200}{800} = 0.125$
- $400: \frac{400-200}{800} = 0.250$
- $600: \frac{600-200}{800} = 0.500$
- $1000: \frac{1000-200}{800} = 1.000$

(b) Z-Score Normalization

Mean (μ) and Standard Deviation (σ)

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = 500$$

$$\sigma = \sqrt{\frac{(200 - 500)^2 + (300 - 500)^2 + (400 - 500)^2 + (600 - 500)^2 + (1000 - 500)^2}{5}}$$

$$\sigma = \sqrt{\frac{90000 + 40000 + 10000 + 10000 + 250000}{5}} = \sqrt{\frac{400000}{5}} = \sqrt{80000} \approx 282.84$$

Calculation ($Z = \frac{X-500}{282.84}$)

- $200: \frac{200-500}{282.84} = -1.061$
- $300: \frac{300-500}{282.84} = -0.707$
- $400: \frac{400-500}{282.84} = -0.354$
- $600: \frac{600-500}{282.84} = 0.354$
- $1000: \frac{1000-500}{282.84} = 1.768$

Q5. Data Transformation in Online Shopping Behavior Analysis

A. Equal-Frequency Binning

Given 10 records $\Rightarrow$ Bin 1 (3 items), Bin 2 (3 items), Bin 3 (4 items).

Bin Assignments

Bin	Assigned Records	Duration Range (s)
Bin 1 (Short)	C1 (15), C2 (30), C3 (45)	15s - 45s
Bin 2 (Medium)	C4 (60), C5 (120), C6 (300)	60s - 300s
Bin 3 (Long)	C7 (600), C8 (900), C9 (1800), C10 (3600)	600s - 3600s

Effectiveness Analysis

Equal-frequency binning handles skewed data well by ensuring equal sample distribution per bin. It prevents long-tail outliers (like 3600s) from creating sparse/empty bins.

B. Z-Score Normalization

i. Mean (μ) & Standard Deviation (σ)

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10} = \mathbf{747\ s}$$

$$\sigma \approx \mathbf{1142.14\ s}$$

ii. Applied Values ($Z = \frac{X-747}{1142.14}$)

Customer_ID	Duration (s)	Z-Score
C1	15	-0.641
C2	30	-0.628
C3	45	-0.615
C4	60	-0.602
C5	120	-0.549
C6	300	-0.391
C7	600	-0.129
C8	900	0.134
C9	1800	0.922
C10	3600	2.498

iii. Handling Extreme Values

Z-score keeps extreme values (C9 = 1800s, C10 = 3600s) bound in standard deviation space without squishing all short visits into an indistinguishable range.

C. Min-Max Normalization [0, 1]

Formula:

$$v' = \frac{v - 15}{3600 - 15} = \frac{v - 15}{3585}$$

Transformed Values Comparison Table

Customer_ID	Raw Duration (s)	Min-Max [0,1]	Z-Score
C1	15	0.0000	-0.641
C2	30	0.0042	-0.628
C3	45	0.0084	-0.615
C4	60	0.0126	-0.602
C5	120	0.0293	-0.549
C6	300	0.0795	-0.391
C7	600	0.1632	-0.129
C8	900	0.2469	+0.134
C9	1800	0.4979	+0.922
C10	3600	1.0000	+2.498

Comparison Summary

- Min-Max Normalization: Strongly affected by the outlier (3600s). As a result, C1 through C5 are compressed into a tiny range (0.000 to 0.029), losing variance between short visits.
- Z-Score Normalization: Better preserves the relative distribution across smaller values while identifying C10 as an outlier ($+2.5\sigma$). For clustering skewed duration data, log transformation + Z-score or Equal-Frequency Binning is preferred over standard Min-Max.